Qingwei Ben

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Education

The Chinese University of Hong Kong, Hong Kong, China Sep. 2024 – Jun. 2028

Ph.D in Information Engineering at Multimedia Lab (MMLab)

- Advised by Prof. Dahua Lin

Tsinghua University, Beijing, China Sep. 2020 – Jun. 2024

B.Sc. in Mathematics and Physics and B.Eng. in Software Engineering

Research Experience

Research Intern, mentored by Dr. Linxi "Jim" Fan and Prof. Yuke Zhu Jul. 2026 – Present

GEAR, Nvidia Research, California, USA

Research Intern, mentored by Dr. Jiangmiao Pang Sep. 2023 – Jun. 2026

OpenRobotLab, Shanghai AI Lab, Beijing & Shanghai, China

Research Assistant, advised by Prof. Huazhe Xu Mar. 2023 – Mar. 2024

TEA Lab, IIIS, Tsinghua University, Beijing, China

Visiting Researcher, advised by Prof. Chuang Gan Jun. 2023 – Sep. 2023

MIT-IBM Watson AI Lab & Umass Amherst, Massachusetts, USA

Representative Publications

(* shows equal contribution)

[1] Gallant: Voxel Grid-based Humanoid Locomotion and Local-navigation across 3-D Constrained Terrains **Qingwei Ben***, Botian Xu*, Kailin Li*, Feiyu Jia, Wentao Zhang, Jingping Wang, Jingbo Wang, Dahua Lin, Jiangmiao Pang. (CVPR 2026 Highlight)

[2] HOMIE: Humanoid Mobile Manipulation with Isomorphic Exoskeleton Cockpit **Qingwei Ben***, Feiyu Jia*, Jia Zeng, Junting Dong, Dahua Lin, Jiangmiao Pang. (RSS 2025, RSS 2025 WCBM Workshop)

Core Projects

(* shows equal contribution, + shows core contributors, # shows project leads.)

[1] GRAIL: Generating Humanoid Loco-Manipulation from 3D Assets and Video Priors Tianyi Xie*, Haotian Zhang*, Jinhyung Park*, Zi Wang*, Bowen Wen, Jiefeng Li, Xueting Li, **Qingwei Ben**, Haoyang Weng, Yufei Ye, David Minor, Tingwu Wang, Chenfanfu Jiang, Sanja Fidler, Jan Kautz, Linxi "Jim" Fan, Yuke Zhu, Zhengyi Luo#, Umar Iqbal#, Ye Yuan#. (Nvidia DAIR, CoRL 2026 Oral)

[2] SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control Zhengyi Luo*, Ye Yuan*, Tingwu Wang*, Chenran Li*, Sirui Chen+, Fernando Castañeda+, Zi-Ang Cao+, Jiefeng Li+, David Minor+, **Qingwei Ben+**, Xingye Da+, Runyu Ding, Cyrus Hogg, Lina Song, Edy Lim, Eugene Jeong, Tairan He, Haoru Xue, Wenli Xiao, Zi Wang, Simon Yuen, Jan Kautz, Yan Chang, Umar Iqbal, Linxi Fan, Yuke Zhu. (Nvidia Gear, Science Robotics)

[3] GRUtopia: Dream General Robots in a City at Scale Hanqing Wang*, Jiahe Chen*, Wensi Huang*, **Qingwei Ben***, Tai Wang*, Boyu Mi*, Tao Huang, Siheng Zhao, Yilun Chen, Sizhe Yang, Peizhou Cao, Wenye Yu, Zichao Ye, Jialun Li, Junfeng Long, Zirui Wang, Huiling Wang, Ying Zhao, Zhongying Tu, Yu Qiao, Dahua Lin, Jiangmiao Pang. (Shanghai AI Lab OpenRobotLab, Technical Report)

Other Publications

(* shows equal contribution, + shows core contributors, # shows project leads.)

[1] Feel Robot Feels: Tactile Feedback Array Glove for Dexterous Manipulation Feiyu Jia*, Xiaojie Niu*, Sizhe Yang*, **Qingwei Ben**, Tao Huang, Feng Zhao, Jingbo Wang, Jiangmiao Pang.

[2] Opening the Sim-to-Real Door for Humanoid Pixel-to-Action Policy Transfer Haoru Xue*, Tairan He*, Zi Wang*, **Qingwei Ben**, Wenli Xiao, Zhengyi Luo, Changliu Liu, Guanya Shi, S. Shankar Sastry, Linxi Fan, Yuke Zhu. (CVPR 2026)

[3] VIRAL: Visual Sim-to-Real at Scale for Humanoid Loco-Manipulation Tairan He*, Zi Wang*, Haoru Xue*, **Qingwei Ben***, Wenli Xiao, Ye Yuan, Zhengyi Luo, S. Shankar Sastry, Changliu Liu, Guanya Shi, Linxi Fan, Yuke Zhu. (CVPR 2026)

[4] One-Policy-Fits-All: Geometry-Aware Action Latents for Cross-Embodiment Manipulation Juncheng Mu, Sizhe Yang, Hojin Bae, Feiyu Jia, **Qingwei Ben**, Boyi Li, Huazhe Xu, Jiangmiao Pang. (ICRA 2026)

[5] TeleOpBench: A Simulator-Centric Benchmark for Dual-Arm Dexterous Teleoperation Hangyu Li*, Qin Zhao*, Haoran Xu, Xinyu Jiang, **Qingwei Ben**, Feiyu Jia, Haoyu Zhao, Liang Xu, Jia Zeng, Hanqing Wang, Bo Dai, Junting Dong, Jiangmiao Pang.

[6] SMAP: Self-supervised Motion Adaptation for Physically Plausible Humanoid Whole-body Control Haoyu Zhao*, Sixu Lin*, **Qingwei Ben**, Minyue Dai, Hao Fei, Jiangmiao Pang, Jingbo Wang, Junting Dong, Hua Zou.

[7] BeamDojo: Learning Agile Humanoid Locomotion on Sparse Footholds Huayi Wang, ZiRui Wang, Junli Ren, **Qingwei Ben**, Tao Huang, Weinan Zhang, Jiangmiao Pang. (RSS 2025)

[8] Learning Humanoid Standing-up Control across Diverse Postures Tao Huang, Junli Ren, Huayi Wang, ZiRui Wang, **Qingwei Ben**, Muning Wen, Xiao Chen, Jianan Li, Jiangmiao Pang. (RSS 2025 Best System Paper Finalist)

[9] VB-Com: Learning Vision-Blind Composite Humanoid Locomotion Against Deficient Perception Junli Ren, Tao Huang, Huayi Wang, ZiRui Wang, **Qingwei Ben**, Jiangmiao Pang, Ping Luo. (ICRA 2026)

[10] RoboDuet: Learning a Cooperative Policy for Whole-body Legged Loco-Manipulation Guoping Pan*, **Qingwei Ben***, Zhecheng Yuan, Guangqi Jiang, Yandong Ji, Shoujie Li, Jiangmiao Pang, Houde Liu, Huazhe Xu. (RA-L+IROS 2025, ICRA 2024 Mobile-Manipulation Workshop)

[11] DiffTactile: A Physical-based Differentiable Tactile Simulation for Contact-rich Robotic Manipulation Zilin Si*, Gu Zhang*, **Qingwei Ben***, Branden Romero, Xian Zhou, Chao Liu, Chuang Gan. (ICLR 2024, RSS 2024 Noosphere Workshop Best Paper)

Skills

- *Professional Skills*: Python, C/C++, Java, Javascript, ROS, Pytorch, Docker, Qt, HTML, CSS, SQL, VUE, Flask, Taichi, SolidWorks, Isaac Sim, Isaac Gym, Mujoco
- *Language*: Madarin (native), English (proficient, TOFEL 107/120), Tongtai subgroup (native)

Research Interests

Embodied AI, Humanoid, Whole-body Control, Loco-Manipulation, World Action Model

Teaching Service

- CUHK IERG 4998: Final Year Project, Autumn 2024 (Teaching Assistant)
- CUHK IERG 4999: Final Year Project, Spring 2025 (Teaching Assistant)
- CUHK IERG 4998: Final Year Project, Autumn 2025 (Teaching Assistant)
- CUHK IERG 4999: Final Year Project, Spring 2026 (Teaching Assistant)

Honors & Results

- RSS 2024 Noosphere Workshop Best Paper (DiffTactile)
- RSS 2025 Best System Paper Finalist (HoST)
- Conference & Journal: RSS×3, CVPR×3, ICLR×1, ICRA×2, CoRL×1, RA-L×1, Science Robotics×1

Community Service

- Reviewer: IROS 2025, RSS 2025 Workshop WCBM, ICRA 2026, TMECH, CVPR 2026, ECCV 2026, RA-L, RSS 2026, IROS 2026, T-RL, T-RO, TVCG, ICRA2027